

Asad Mahmood

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Education

National University of Sciences and Technology (NUST)

Nov. 2021– May 2025

Bachelor of Civil Engineering

Islamabad, Pakistan

• **CGPA:** 3.44 / 4.00

Relevant Coursework

- | | | | |
|-----------------------|------------------------|------------------------|-------------------------|
| • Structural Analysis | • Environmental Eng | • Construction Project | • Construction Project |
| • Fluid Mechanics | • Computer Aided Civil | Management | Scheduling |
| • Plain & Reinforced | Engineering | • Soil Mechanics | • Construction Contract |
| Concrete | • Steel Structure | • AI in Construction | Management |

Experience

Nexus Construction Technology

June 2024 – August 2024

Intern

Islamabad, Pakistan

- Optimized concrete mixes for a 3D printer to improve printability and structural performance.
- Developed a Python and machine learning-based tool with a user interface for Pakistan's first 3D concrete printer.

Marina Sports City

June 2023 – August 2023

Intern

Lahore, Pakistan

- Assisted in surveying, layout, and construction of roads and multistorey buildings.
- Observed raft foundation construction and ensured structural design compliance on-site.

The Citizens Foundation

May 2024 – June 2024

Senior Volunteer

Islamabad, Pakistan

- Led a team of 10 to teach core subjects and organize extracurricular activities for underprivileged students in grades 3–5.

Projects

Final Year Design Project – Bricklaying Robotic Arm | BIM, Dynamo, Revit

- Developed an innovative robotic arm for bricklaying using BIM tools like Dynamo and Revit.
- Aimed to improve construction efficiency and accuracy while addressing labor shortages and safety issues in Pakistan.

Rooftop Café Renovation | Revit, Primavera

- Redesigned the department's cafeteria into a rooftop garden space.
- Created detailed 3D models in Revit, project schedules in Primavera and detailed BOQs with accurate cost estimates.

Concrete Using Recycled Aggregate | Mix Design, Technical Report

- Designed M40 grade concrete incorporating recycled aggregates to promote sustainable construction practices.
- Analyzed compressive strength under pond and ambient curing, and compiled results in a technical report.

3D Modelling and Simulation of Oval-Shaped School | AutoCAD

- Utilized AutoCAD to draft floor plans and structural designs for a 3D oval-shaped building inspired by Rajkumari Ratnavati School.
- Designed structural elements including columns, beams, foundations, slabs, and MEP systems.

Construction Contracts Management Simulation – FIDIC 2017 | Contractual Simulation

- Collaborated on an EOT claim under FIDIC 2017, citing justified delays and applying relevant clauses.
- Reviewed peer claims and successfully defended ours in arbitration, achieving the highest project grade.

Designing & Cost Estimation of Residential House

- Conducted quantity surveying and cost estimation for a two-story house, covering all phases from excavation to finishing.
- Designed structural elements (beams, columns and foundation) with safety compliance.

Technical Skills

Programming Languages: Python, C++, MATLAB

Software & Tools: AutoCAD, Revit, ArcGIS, Primavera P6, Bluebeam, MS CAD, Microsoft Office, Excel, Canva

Leadership / Extracurricular

NUST Digital Club

August 2024 – May 2025

Treasurer

NUST

- Led financial planning and expense tracking for club activities and annual events.
- Managed cross-functional teams including Finance, Decor, Logistics, and Event Management.

NICE Student Council

August 2023 – May 2024

JS to Treasurer

NUST

- Oversaw budgeting, financial management, and equipment procurement for various student-led initiatives.
- Supported coordination across departments to ensure smooth execution of council operations.